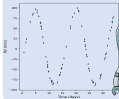


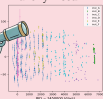
Synthetic tasks

100 tasks,
3 difficulty tiers,
controlled noise &
orbital configurations



Real data tasks

20 tasks from
published RV
datasets, multi-
instrument,
anonymised

Planet
MultiplicityPeriod
Coverage

Signal-to-Noise Ratio



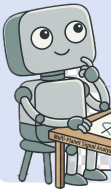
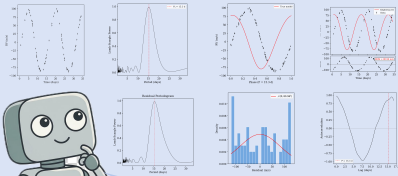
N=12
of
Observations



Resonant Configuration



Correlated Noise



Pondering... *Stargazing!* ✨
It's time to submit my
answers and get feedback!

Statistical & Physical criteria for exoplanet candidates



ok_delta_bic
Model Complexity



ok_match
Parameters matching



ok_rms
Root mean square



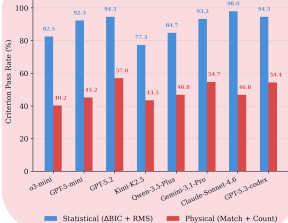
ok_count
Number of planets



Good statistics ≠ Good physics!

Model	Statistical (Δ BCI + RMS) (%)	Physical (Match + Count) (%)
g3-mini	82.5	40.2
GPT-3-mini	92.3	45.2
GPT-5.2	94.3	57.0
Kimi-K2.5	77.3	43.5
Qwen-3.5-Plus	84.7	46.8
Gemini-3.1-Pro	93.2	54.7
Claude-Sonnet-4.0	98.0	46.8
GPT-5.3-codex	94.3	54.1

Legend: Blue bars represent Statistical (Δ BCI + RMS), Red bars represent Physical (Match + Count).



I burned nearly **1M tokens** and got a **wrong** answer.

